

SREE SANKARAN CHACKOTH

APPLIED ML & AGENTIC AI ENGINEER

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PROFESSIONAL SUMMARY

Applied ML and Agentic AI Engineer with 3+ years of experience building production-oriented computer vision, machine learning, and LLM systems across academic, government, and industry settings. Currently leading an industry-partnered applied AI project at Lambton College, owning model development, synthetic data generation, OCR, benchmarking, error analysis, and cloud inference for automated venue-map understanding. Builds agentic systems with LangGraph/LangChain, Microsoft Foundry and Agent Framework, RAG, MCP/tool integrations, evaluation, and workflow automation. Strong Python, PyTorch, TensorFlow/Keras, OpenCV, and MLOps foundation.

TECHNICAL SKILLS

Machine Learning & Computer Vision: Python, PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, YOLO, object detection, multi-class classification, OCR, synthetic data generation, feature engineering, model evaluation, benchmarking, time-series forecasting, LightGBM, XGBoost

Agentic AI & Generative AI: LangGraph, LangChain, Microsoft Foundry, Microsoft Agent Framework, RAG, multi-agent orchestration, MCP/tool integrations, Hugging Face, Mistral, LLM fine-tuning/pre-training, evaluation, guardrails, n8n automation

MLOps, Cloud & Data: Lightning AI, AWS (EC2, Lambda, SageMaker, CodeDeploy), Docker, Kubernetes, GitHub Actions, Terraform, MLflow, REST APIs, SQL, PySpark, Databricks, Airflow, Power BI

PROFESSIONAL EXPERIENCE

AI Research Technician / AI Specialist | Lambton College × EventLinx

Feb 2026 - Present

Sarnia, ON

- Own the end-to-end ML pipeline for automated venue-map understanding, converting SVG, PDF, PNG, and JPG layouts into structured spatial data for seats, sections, stages, exits, tables, and accessibility features using computer vision, deep learning, and OCR.
- Developed and trained multi-class seat detection/classification workflows and a synthetic-data generation pipeline that produces large-scale annotated training examples for regular, wheelchair, reserved, and hearing-accessible seating.
- Benchmarked classical CV, segmentation, and YOLO-based approaches; implemented OCR-boundary handling and seat-number parsing to reduce stage over-prediction, tiny-object errors, and conflicts between visual detections and embedded numbering.
- Engineered testable inference components and structured output contracts for downstream applications, with systematic model evaluation, regression testing, error analysis, and architecture documentation across the research lifecycle.
- Deployed and iterated on GPU-backed training/inference workflows with Lightning AI while working directly with the industry partner, developers, and research coordinators to translate product requirements into measurable ML experiments and client-facing deliverables.

AI Research Assistant | Applied Research Dept., Algonquin College

Apr 2025 - Dec 2025

Ottawa, ON

- Led technical research and proposal development for Horizon Europe-aligned AI initiatives in emergency medical services and privacy-preserving analytics; defined ML pilot architectures, data requirements, and evaluation plans.
- Coordinated 5+ government/industry stakeholders and authored funding proposals across Mitacs, NSERC ARD, Horizon Europe, and institutional programs, contributing to an initial \$150K research funding commitment.

ML Engineer - Forecasting Systems (Capstone) | Immigration, Refugees & Citizenship

Sep 2024 - Apr 2025

Canada

Ottawa, ON

- Built a production-oriented time-series forecasting pipeline across 15+ Canadian immigration categories; processed 2M+ records with scalable Python ETL and maintained 99.5% data quality.
- Implemented and evaluated Prophet, ARIMA, and Exponential Smoothing models, reaching 85%+ forecast accuracy across multiple horizons for planning scenarios affecting 400,000+ annual applicants.

AI/ML Engineer & Team Lead (Capstone) | Communications Security Establishment Canada

Sep 2024 - Apr 2025

Ottawa, ON

- Led a team building a production-ready Windows executable threat-classification CLI capable of processing 10,000+ files/hour; owned feature engineering, model training, evaluation, and delivery.
- Trained and benchmarked Logistic Regression, Random Forest, XGBoost, and LightGBM over 8M+ records; final LightGBM model achieved 63% accuracy on a highly imbalanced malware-classification problem.

Python Application Developer | Infidata Technologies

Mar 2022 - Jun 2022

Bengaluru, India

- Built an event-driven ETL pipeline with AWS Lambda and RDS, improving data freshness to hourly and eliminating an estimated 40 hours/month of manual QA; implemented blue-green deployments through AWS CodeDeploy.

SELECTED AGENTIC AI & ML PROJECTS

Agents League - Battle #2: Reasoning Agents with Microsoft Foundry Challenge Project	2026
• Built a multi-agent reasoning prototype with Python, Microsoft Foundry, and Microsoft Agent Framework, orchestrating specialized agent roles with external tools/APIs and MCP-compatible integrations for a demoable end-to-end workflow. • Structured the solution around Microsoft IQ capabilities and incorporated synthetic test data, telemetry/evaluation, and guardrail-oriented validation to make agent behavior observable, repeatable, and easier to assess.	
AI for Regulatory Compliance RAG, LangGraph, Multi-Agent Systems, Jira	
• Designed a LangGraph-based multi-agent RAG system that retrieves regulatory evidence, reasons over internal processes, detects compliance gaps, and automatically creates or updates Jira tickets, translating complex legal requirements into engineering action.	
Customer Churn Intelligence Agent LangChain, Mistral, BERT, MLOps	
• Combined transaction features with BERT-based sentiment signals in a churn model that achieved 85% AUC, then used a LangChain agent with Mistral to generate targeted retention actions from model outputs.	

EDUCATION

Post-Graduate Certificate, Business Intelligence Systems Infrastructure Algonquin College, Ottawa, ON	2025
Post-Graduate Certificate, Artificial Intelligence Software Development Algonquin College, Ottawa, ON	2024
B.Tech, Electronics & Communication Engineering Presidency University, Bengaluru, India	2022

CERTIFICATIONS & COMMUNITY

Certifications: AI Applications with Azure (LinkedIn Learning), Generative AI with OpenAI (DeepLearning.AI), DevOps Foundations Series (LinkedIn Learning).

Technical Community: Public AI/MLOps writing with 5,000+ monthly readers on multi-agent workflows, model quantization, and applied ML; administrator of a 600+ member AI-careers Discord; open-source contributor on GitHub and Hugging Face; member of the Google Cloud Developers Community, Ottawa.